An end-to-end normal-crossing example: the model $N(x_0x_1, 1)$

Claude, with Daniel Murfet

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Abstract

Unit 218 follows Astra #24's top recommendation (programme H2-lite): after the hand-off, demonstrate applicability with one bounded example rather than enlarge the theorem catalogue. Observations $y_1, \ldots, y_n$ are modelled as $N(x_0x_1, 1)$ with $x \in (-1, 1]^2$ and a continuous prior density ρ; against the truth $N(0, 1)$ the likelihood ratio is *exactly* $\exp(-\frac{n}{2}(x_0x_1)^2 + \sqrt{n}\, Z_n x_0 x_1)$ with $Z_n = n^{-1/2} \sum_i y_i$, i.e. the grammar paper's dressed normal-crossing kernel at $\beta = \frac{1}{2}$, $N = \sqrt{n}$, $h = 0$, $k = (1, 1)$, phase $2Z_n$ (`ncLikelihood_eq`). No resolution of singularities, no likelihood remainder and no empirical-process limit is assumed: the quadrant decomposition is Headline XIX's `symBox_phase_eq_sum` and the phase is a datum. The fluctuation variable is $\sqrt{n}\, x_0x_1$; its posterior moment generating function is a ratio of two model integrals with phases $2Z_n + 2\theta$ and $2Z_n$ (exponential tilt = phase shift), and the end-to-end theorem is

$$\mathbb{E}_{\text{post}}\left[e^{\theta\sqrt{n}\, x_0x_1}\right] \longrightarrow e^{z\theta + \theta^2/2} \qquad \text{whenever } Z_n \to z$$

(`ncFluctMGF_tendsto`): the posterior law of $\sqrt{n}\, x_0x_1$ is asymptotically $N(Z_n, 1)$, centred at the empirical phase. Only $\rho(0) > 0$ is needed. Zero `sorry`/`axiom`; 9095 jobs.

1 The things formalised

```
theorem integral_quadKernel (  a) :   s, quadKernel   a s = √( / ) * exp (  a²/4)
theorem phaseMoment_one_add_neg : phaseMoment   1 a + phaseMoment   1 (-
a) = √( / ) e^{ a²/4}
theorem chart_const_phase_tendsto_moving : c  → c , N  → ∞
  chartIntegral … (N ) (fun _ => c )   / leadScale … (N ) → phaseCoeff … (fun _ => c )
def ncLikelihood n y x :=   i, gaussianPDFReal (x 0 * x 1) 1 (y i);  ncPhase n y := (  y)/√n
def ncKernel N z x := exp (-(N²(x x )²)/2 + N z x x );  ncPosteriorMean; ncFluctMGF
theorem ncLikelihood_eq (hn : 0 < n) : ncLikelihood n y x =
  (  i, gaussianPDFReal 0 1 (y i)) * ncKernel (√n) (ncPhase n y) x
theorem ncPosteriorMean_eq : posterior mean = ratio of kernel integrals (truth factor cancels)
theorem nc_integral_tendsto : ( _{symBox 2}   K_{N ,z }) / leadScale … → 2 (0)√(2 ) e^{z ²/2}
theorem ncFluctMGF_tendsto (h 0 : 0 <   0) (hz : ncPhase n (y n) → z) :
  ncFluctMGF n (y n)    → exp (z   +  ²/2)
```

2 Mathematics

Three ingredients on top of the existing block. (i) The *moving-phase* form of the leading-term theorem: with $c_n \to c_0$ and $N_n \to \infty$ the eventual Lipschitz dependence of the normalised chart integral on the phase (unit 209) and the fixed-phase limit (Headline XIII) give $F_{N_n}(c_n) \to \mathcal{C}(c_0)$ — needed because the empirical phase moves with n. (ii) Completing the square:

$J_1(a) + J_1(-a) = \int_{\mathbb{R}} e^{-\beta s^2+\beta a s}\, ds = \sqrt{\pi/\beta}\, e^{\beta a^2/4}$, so the four quadrants ($\varepsilon_\sigma = \pm 1$ twice each) sum to $2\rho(0)\sqrt{2\pi}\, e^{z^2/2}$ at $\beta = \frac{1}{2}$, phase $2z$, and the ratio of the numerator (phase $2z + 2\theta$) to the denominator is $e^{((z+\theta)^2-z^2)/2} = e^{z\theta+\theta^2/2}$. (iii) The exact likelihood identity, an elementary Gaussian computation. The example sits in the $m = 2$ logarithmic regime: the scale is $N^{-1}\log N$ and finite-n corrections are $O(1/\log n)$; a numerical check ($\rho = 1+0.3x_0+0.2x_1^2$, $z = 0.7$, $\theta = 0.5$, target 1.6080) gives $1.407, 1.498, 1.532, 1.559$ at $n = 10^2, 10^4, 10^6, 10^8$ with gap$\times \log n \approx 1$ throughout, consistent with the theorem and with a $C/\log n$ correction.

3 Lean notes

- `integral_sub_right_eq_self` needs `(  := volume)` or the `IsAddRightInvariant` instance search is stuck on a metavariable.
- `Tendsto.div` produces a `Pi.div` of lambdas; before rewriting the pointwise identity, `simp only [Pi.div_apply]`. Pointwise goals from `integral_congr_ae` again need `beta_reduce` before `rw`.
- Sums over σ : Fin 2 $\to$ Bool: `rw [← (piFinTwoEquiv fun _ => Bool).symm.sum_comp, Fintype.sum_prod_type]; simp only [Fintype.sum_bool, piFinTwoEquiv, Fin.cons_zero, Fin.cons_one, …]` enumerates the four quadrants explicitly.
- The `ring` fallback message "Try this: ring_nf" is an info, not a warning, but the build log shows it; use `ring_nf` where `ring` only normalises.

4 Status

Headline XX (end-to-end normal-crossing example) added to the index. Next (unit 219): the Gaussian-data layer — for y_i i.i.d. $N(0,1)$ the phase Z_n has variance one, so by Chebyshev and uniform convergence on compact phase sets the difference $\mathbb{E}_{\text{post}}[e^{\theta\sqrt{n}x_0x_1}] - e^{Z_n\theta+\theta^2/2}$ tends to zero in probability.